

Hydraulic Jump

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Hydraulic jump is an important phenomenon occurring in open flow channels and its analysis is an important part in engineering. In this paper, we will experimentally find the head losses and conjugate depth and verify it using theoretical equations available.

Nomenclature

Q = Flow rate
 y = Height of the channel
 b = width of the channel
 y_c = critical depth
 v = velocity of the flow
 h_L = energy loss
 Fr = Froude's number
 y_1 = height of flow before the jump
 y_2 = height of flow after the jump

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I. Introduction

Hydraulic jumps are phenomenon occurring in open channels such as rivers, spill ways etc. When the flowing fluid interacts with surrounding objects, the flow kinetic energy is transformed to potential energy, or in other words, when fluid travels from a high velocity region to a low velocity region, its kinetic energy is transformed to potential energy which leads to the increase in height of fluid flow. A figure representing hydraulic jump is shown in figure 1.

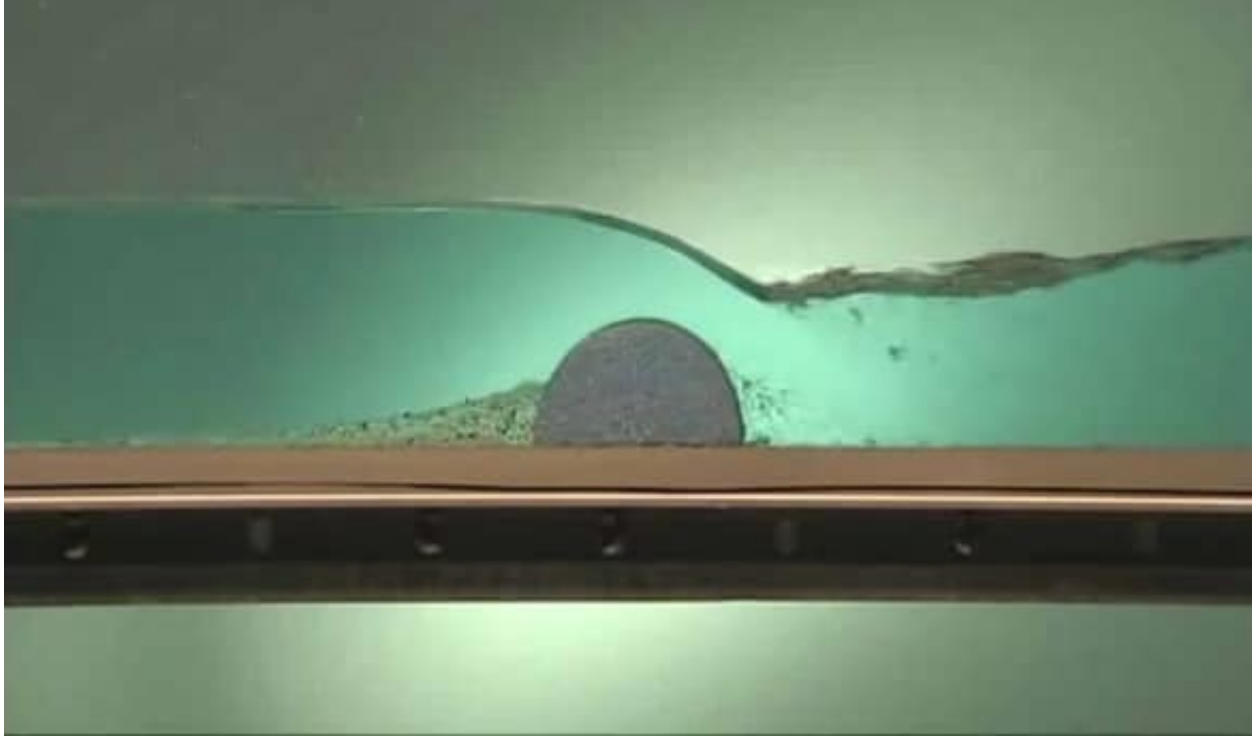


Figure 1. hydraulic jump

Hydraulic jump occurs for dissipating the energy in fluid flow so that the scouring of hydraulic structure can be prevented. They are also used in mixing chemicals. Hydraulic jump can be classified based on the Froude's number(Fr):

1. Undular Hydraulic Jump ($1 < Fr < 3$)
2. Weak Jump ($3 < Fr < 6$)
3. Oscillating Hydraulic Jump ($6 < Fr < 20$)
4. Steady Hydraulic Jump ($20 < Fr < 80$)
5. Strong Hydraulic Jump ($Fr > 80$)

As stated, energy dissipation occurs at hydraulic jump. So energy analysis is not easy to carry out. So we use linear momentum equation to do the analysis. In this experiment, water is allowed to flow in the open channel and required parameters are measured. Using those parameters, head losses are calculated, and experimental data is compared with those obtained theoretically.

II. Theory

A hydraulic jump is observed in a flow in a rectangular channel, when the flow transitions from super-critical to sub-critical flow. Super-critical flow is the flow in which Froude Number is greater than 1 and

sub-critical flow is the flow in which Froude Number is less than 1. The Froude Number for a flow in a rectangular channel is given by:

$$Fr = \frac{v}{\sqrt{gd}}$$

For the theoretical analysis of hydraulic jumps, the flow can be considered to have a hydro-static pressure distribution, and also the flow to be incompressible, inviscid and uniform.

Consider a hydraulic jump in an open channel as the following:

*****place Figure*****

Let 1 and 2 represent the upstream and downstream section. Since the width of the channel (w) is constant, the continuity equation for incompressible flow ($Q_1 + Q_2$) can be written as:

$$v_1 y_1 = v_2 y_2$$

By applying conservation of linear momentum along the x-direction, we can write the following expression:

$$\sum F_x = F_1 - F_2 = \rho g \frac{y_1^2}{2} - \rho g \frac{y_2^2}{2} = Q\rho(v_2 - v_1)$$

Since, $Q = v_1 y_1 = v_2 y_2$, the above equation can be written as:

$$\frac{Q}{g} \left(\frac{Q}{y_2} - \frac{Q}{y_1} \right) = \frac{1}{2} (y_1^2 - y_2^2) = \frac{1}{2} (y_1 - y_2) (y_1 + y_2)$$

$$\frac{v_1^2 y_1^2}{g} \frac{\left(\frac{1}{y_2} - \frac{1}{y_1} \right)}{y_1 - y_2} = \frac{1}{2} (y_1 + y_2) = \frac{v_1^2 y_1^2}{g} \frac{1}{y_1 y_2}$$

$$\frac{v_1^2}{g} = \frac{1}{2} \frac{y_2}{y_1} [y_1 + y_2]$$

$$\frac{v_1^2}{g y_1} = \frac{1}{2} \left[\frac{y_2}{y_1} + \frac{y_2^2}{y_1^2} \right]$$

Since y_1 is the depth of the super-critical flow (d), the above equation can be represented in terms of the Froude Number as:

$$\frac{v_1^2}{g y_1} = \frac{v_1^2}{g d} = \frac{1}{2} \left[\frac{y_2}{y_1} + \frac{y_2^2}{y_1^2} \right] = Fr^2$$

$$8Fr^2 = 4 \left[\frac{y_2^2}{y_1^2} + \frac{y_2}{y_1} \right]$$

Finding the roots of the above equation, we obtain:

$$\frac{y_2}{y_1} = \frac{1}{2} \left[\sqrt{8Fr^2 + 1} - 1 \right]$$

Hence it can be observed from the theoretical analysis that, the jump is dependant only on the Froude Number of the flow.

Since the specific energy at of the flow at both the sections considered can be expressed as:

$$E_{s1} = y_1 + \frac{v_1^2}{2g} \text{ and } E_{s2} = y_2 + \frac{v_2^2}{2g}$$

The loss in specific energy can be calculated as $E_1 - E_2$:

$$E_{s1} - E_{s2} = \left[y_1 + \frac{v_1^2}{2g} \right] - \left[y_2 + \frac{v_2^2}{2g} \right] = (y_1 - y_2) + \frac{q^2}{2g} \left[\frac{1}{y_1^2} - \frac{1}{y_2^2} \right]$$

Which can be simplified into:

$$E_{sLoss} = \frac{(y_2 - y_1)^3}{4y_1y_2}$$

Hence the theoretical specific energy loss is obtained as a function of the upstream and downstream conjugate depths.

III. Experimental Setup

- The figure 2 shows the overall setup of the experiment. It has a wooden frame as base over which a thermocole base it kept. The length, width and height of the channel is 1m, 3cm and 6cm respectively. The whole frame is inclined at an angle of 10° .



Figure 2. overall setup

- Figure 3 shows the sluice gate which can be lifted for the flow to pass and the height of gate can be adjusted to alter the flow rate. Each division in the gate is of 5mm.

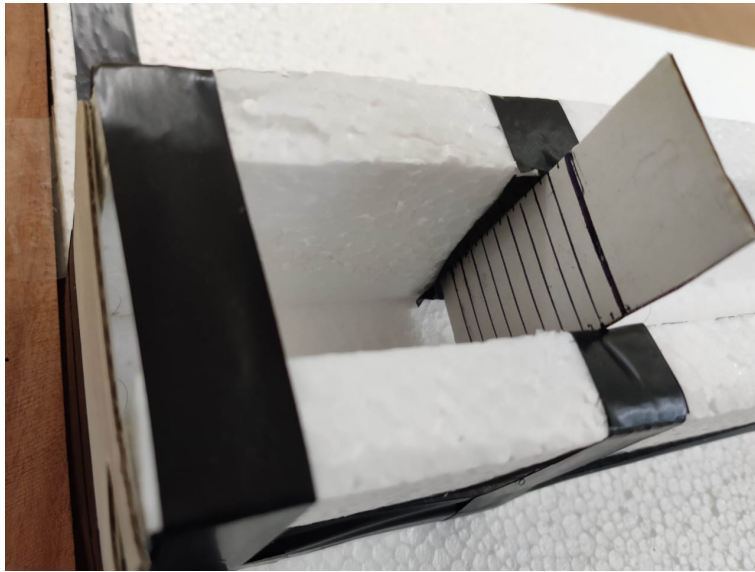


Figure 3. sluice gate

- The water is taken from a tap and is fed into the reservoir tank using pipe connections as shown in Figure 4. The volume flow rate from the pipe was kept at 0.166 L/s.



Figure 4. Pipe connection

- Figure 5 shows a plate which is used to block the flow so that jump can be observed. It is kept at the end of the channel.

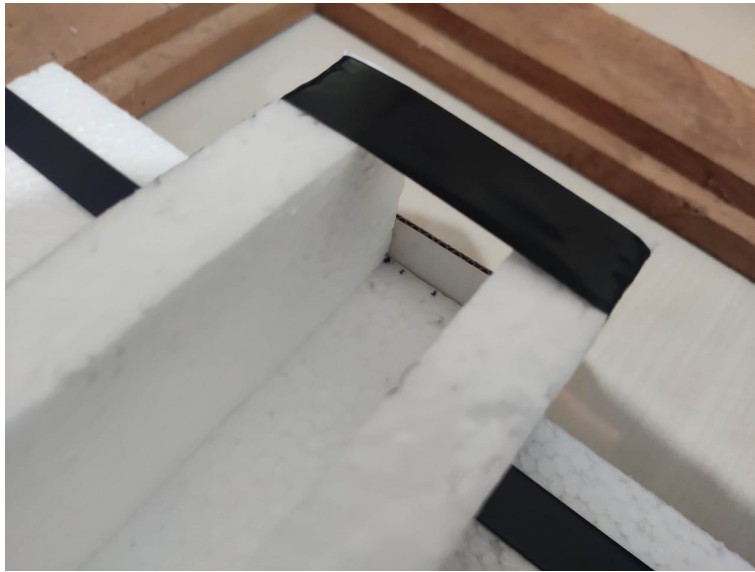


Figure 5. plate used to block the flow

IV. Results and Discussions

*****place all graphs*****

V. Procedure

- A channel with 30 mm width and 1.5 m length is taken, with a pipe attached to the reservoir tank with a maximum flow rate of 0.1666 L/s.
- Sluice gate is raised so that the opening is initially at 5mm. End of the channel is blocked with a plate so that the jump is obtained.
- After the jump is obtained, its height is measured using scale.
- This set of procedure is repeated for various heights ranging from 5mm to 14 mm, and the corresponding heights of the jump is measured.

VI. Conclusion

Appendix A: Observation Tables

Table 1. The Experimental and Theoretical Value of Conjugate depth obtained

Sl. No	Depth of Gate y_1 (mm)	Experimental Depth of Jump Measured (y_2) (mm)	Velocity at Super-Critical Flow Upstream (v_1) (m/s)	Froude Number	Experimental Conjugate Depth ($\frac{y_2}{y_1}$)	Theoretical Conjugate Depth ($\frac{y_2}{y_1}$)	Error % in Theoretical and Experimental Conjugate Depth
1	5.00	30.00	1.07	4.82	6.00	6.33	5.26
2	7.50	22.00	0.71	2.62	2.93	3.24	9.55
3	10.00	18.00	0.53	1.70	1.80	1.96	8.20
4	12.50	14.00	0.43	1.22	1.12	1.30	13.52

Table 2. The Experimental and Theoretical Value of Specific Energy Loss obtained

Sl. No	Depth of Gate y_1 (mm)	Froude Number	Specific Energy Loss calculated with experimental values(m)	Specific Energy Loss calculated with theoretical values(m)	%Error in Experimental and Theoretical Energy Loss values Calculated
1	5.00	4.82	0.0260417	0.02994	13.01
2	7.50	2.62	0.0046191	0.00652	29.20
3	10.00	1.70	0.0007111	0.00113	37.10
4	12.50	1.22	0.0000048	0.00006	92.22

Table 3. Experimental and Theoretical Values of % Energy Dissipation

Sl. No	Depth of Gate y_1 (mm)	Froude Number	% Energy Dissipation (Theoretical)	% Energy Dissipation (Experimental)
1	5.00	4.82	47.48224309	41.30327279
2	7.50	2.62	19.59268868	13.87127861
3	10.00	1.70	4.612031402	2.901017576
4	12.50	1.22	0.284430495	0.022128844

Table 4. Experimental and Theoretical Values of % Energy Dissipation

Sl. No	Depth of Gate y_1 (mm)	Froude Number	% Energy Dissipation (Theoretical)	% Energy Dissipation (Experimental)	% Difference between Theoretical and Experimental Energy Dissipation
1	5.00	4.82	47.48224309	41.30327279	13.01
2	7.50	2.62	19.59268868	13.87127861	29.20
3	10.00	1.70	4.612031402	2.901017576	37.10
4	12.50	1.22	0.284430495	0.022128844	92.22

Appendix B1: Sample Calculations

Here the maximum flow rate $Q = 0.1666 \text{ L/s}$.

Dimensions of the channel :

Height = y (height of the fluid flowing)

Width = b

$$\text{Velocity of flow} = \frac{Q}{A} = \frac{Q}{y_c b}$$

$$\text{Critical depth, } y_c = \frac{v_c}{\sqrt{g y_c}}$$

$$y_c^{3/2} = \frac{Q}{b\sqrt{g}}$$

For $b = 30 \text{ mm}$

$$y_c = 14.6492 \text{ mm}$$

Sluice gate opening height should be anywhere between 0 to 14.6492 mm to obtain the jump.

Now, Let's assume $y_1 = 5 \text{ mm}$.

$$v_1 = \frac{Q}{y_1 b} = 1.1107 \text{ m/s}$$

$$\text{Froude Number, } Fr = \frac{v_1}{\sqrt{g y_1}} = 5.0175$$

Based on the Froude number obtained Jump obtained will be steady jump.

Theoretical jump obtained, $\frac{y_2}{y_1}$:

$$\frac{y_2}{y_1} = \frac{1}{2}[\sqrt{8Fr^2 + 1} - 1]$$

$$\frac{y_2}{y_1} = 6.613383.$$

Experimental jump obtained, $\frac{y_2}{y_1}$:

$$y_2 = 30 \text{ mm.}$$

$$y_1 = 5 \text{ mm.}$$

$$\frac{y_2}{y_1} = 6.$$

Hence experimental and theoretical values are close.

Theoretical, $y_1 = 5 \text{ mm}$ and $y_2 = 33.07 \text{ mm}$

$$\text{Theoretical Energy Loss, } h_L = \frac{(y_2 - y_1)^3}{4y_1 y_2} = 0.03343 \text{ m}$$

Experimental, $y_1 = 5 \text{ mm}$ and $y_2 = 30 \text{ mm}$.

$$\text{Experimental Energy Loss, } h_L = \frac{(y_2 - y_1)^3}{4y_1 y_2} = 0.02604 \text{ m}$$

Appendix C: Error Analysis

The error in the calculated values are obtained and are tabulated below:

SL NO	RPM	Drag (N)	Cd	Error in Drag	Error in Cd
1	0	0			
2	450	3.025139	7.400456	0.009879	2.041472
3	550	4.679242	7.406829	0.009879	1.475829
4	650	6.719019	7.533562	0.009879	1.18818
5	750	9.067163	7.624778	0.009879	1.005448
6	850	11.8487	7.591491	0.009879	0.856132
7	950	14.90847	7.42925	0.009879	0.732738

The sample error calculation for Sl.NO 2 (RPM=450) in the drag and Coefficient of Drag calculation is given below:

The following uncertainties were considered for Error Calculation:

$$\alpha = 1^\circ$$

$$\delta H_t = 0.001m$$

$$\delta H_s = 0.001m$$

$$\delta T = 1K$$

$$\delta V = 0.001mV$$

A. Error analysis in velocity

$$v = \sqrt{\frac{2\rho_e g(H_s - H_t)\sin\alpha RT}{p_o}} = 8.61 \text{ m/s}$$

$$\epsilon_{v,\alpha} = \frac{\partial v}{\partial \alpha} \delta \alpha = \sqrt{\frac{2\rho_e g(H_s - H_t)RT}{4p_o \sin\alpha}} \cos\alpha \delta \alpha = 0.126 \text{ m/s}$$

$$\epsilon_{v,H_t} = \frac{\partial v}{\partial H_t} \delta H_t = -\sqrt{\frac{2\rho_e g \sin\alpha RT}{p_o(H_s - H_t)}} \delta H_t = 0.77 \text{ m/s}$$

$$\epsilon_{v,H_s} = \frac{\partial v}{\partial H_s} \delta H_s = \sqrt{\frac{2\rho_e g \sin\alpha RT}{p_o(H_s - H_t)}} \delta H_s = 0.77 \text{ m/s}$$

$$\epsilon_{v,T} = \frac{\partial v}{\partial T} \delta T = \sqrt{\frac{2\rho_e g(H_s - H_t)\sin\alpha R}{4p_o T}} \delta T = 0.045 \text{ m/s}$$

$$\epsilon_v = \sqrt{\epsilon_{v,\alpha}^2 + \epsilon_{v,H_t}^2 + \epsilon_{v,H_s}^2 + \epsilon_{v,p_o}^2 + \epsilon_{v,T}^2} = 1.2 \text{ m/s}$$

B. Error in dynamic back pressure

$$q = \frac{1}{2}\rho_o v^2 = 43.1 \text{ Pa}$$

$$\epsilon_{qv} = \rho_o v \epsilon_v = 12.03 \text{ Pa}$$

C. Error in Drag (due to least count of Voltage)

$$D = \frac{K_{22}\Delta V_1 - K_{12}\Delta V_2}{K_{22}K_{11} - K_{12}K_{21}} = 3.025 \text{ N}$$

$$\epsilon_D = \sqrt{\left(\frac{K_{22}\delta V_2}{K_{22}K_{11} - K_{12}K_{21}}\right)^2 + \left(\frac{K_{12}\delta V_1}{K_{22}K_{11} - K_{12}K_{21}}\right)^2} = 0.00987 \text{ N}$$

D. Error in C_d

$$C_d = \frac{D}{qS} = 7.4$$

$$\epsilon_{C_d, D} = \frac{\delta D}{qS}$$

$$\epsilon_{C_d, q} = -\frac{D}{q^2 S} \epsilon_q$$

$$\epsilon_{C_d} = \sqrt{\epsilon_{C_d, D}^2 + \epsilon_{C_d, q}^2} = 2.04$$

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References

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